Andrew W. Shenstone

Shenstone, Andrew - Application for Software Engineer, Platform @ Dagster Labs

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SUMMARY

Senior software engineer with 12+ years shipping production systems where reliability, APIs, and data shape the product: B2B SaaS platforms, commerce integrations at scale, and a founder-led application on PostgreSQL with sophisticated modeling and background workflows. I enjoy the builder side of platform work—design, implementation, and refinement—not ticket-chasing ops for its own sake—and I raise quality through reviews, tests, and clear interfaces. Dagster's stack centers Python while valuing TypeScript, React, and GraphQL across the product; Python is not my primary résumé language today, but I am motivated to ramp where your codebase demands it and to contribute immediately on API contracts, Postgres discipline, cross-functional delivery, and pragmatic scalability.

SKILLS

Backend & data systems: PostgreSQL (complex schemas, performance-minded queries), API design & third-party integrations, Node.js, ETL-style pipelines & async workflows (Inngest, webhooks, jobs), GraphQL APIs consumed across clients (Rails GraphQL at Bloom)

Platform & delivery: Monorepos (pnpm, Turbo); scalable module boundaries, CI/CD, automated testing (Vitest, Cypress), AWS (S3, EC2); observability-minded shipping, High-traffic B2B/B2C production systems

Product engineering (TypeScript stack): TypeScript, React, Next.js, Prisma, Stripe, auth & multi-tenant patterns (Clerk, Supabase RLS)

AI/ML: OpenAI API, AI SDK, Vector Embeddings, Natural Language Processing, Retrieval Augmented Generation (RAG)

EXPERIENCE

Founder & Lead Engineer

Plantfolio | November 2023 - Present

Product Leadership & Architecture

- Architected and built a comprehensive plant management platform from the ground up, serving as both technical lead and product owner
- Owned the full feature cycle: joined customer calls to identify friction points, architected solutions, implemented, tested, and verified success in production
- Designed and implemented a sophisticated monorepo architecture with 12+ packages, enabling rapid development and maintainable codebases
- Led product strategy and user experience decisions, making product micro-decisions and navigating ambiguity to deliver value iteratively

Technical Achievements

- AI Integration: Built advanced AI-powered features including plant identification from photos, contextual chatbot (Sage), and smart text extraction from plant tags using OpenAI's vision and language models
- Modern Frontend Architecture: Implemented Next.js with React, React Server Components, and TypeScript, achieving exceptional performance and developer experience
- Complex Data Modeling: Designed and implemented sophisticated PostgreSQL schemas with Prisma, handling complex relationships between users, plants, folios, and media with 15+ interconnected models
- Supabase to Neon: Shipped on Supabase (PostgreSQL with Row-Level Security policies for multi-tenant isolation) before migrating the data layer to Neon; strong hands-on experience with RLS and Postgres-centric access control
- Content Management: Created sophisticated content management system with rich text editing, plant tagging, and flexible data structures

Scale & Performance

- Managed a database of 40,000+ plant species with advanced search capabilities using Algolia and vector embeddings
- Implemented comprehensive media management system with AWS S3, supporting image optimization, galleries, and metadata extraction
- Built robust authentication and authorization systems with Clerk, supporting multiple user types and subscription tiers
- Optimized application performance through React Server Components, efficient data fetching, and strategic caching

Client Case Study - AI Signage

- Delivered a production rollout for an independent garden center to automate plant signage generation; one independent retail customer in ongoing production who has been an enthusiastic reference
- Enabled bulk plant import (plain text) to printable signs in seconds; reduced a weekend-long manual process to minutes after initial setup
- Demonstrated ability to translate customer requirements into scalable technical solutions and shorten feedback loops

Full-Stack Ownership

- Developed complete payment integration with Stripe, subscription management, and user onboarding flows
- Built comprehensive email system with React Email, supporting transactional emails, notifications, and marketing campaigns
- Implemented advanced search and filtering capabilities with Algolia, supporting complex plant queries and user preferences

Code Quality & Leadership

- Established coding standards and architectural patterns that enabled rapid, consistent development
- Implemented comprehensive testing strategy with Vitest and Cypress, maintaining high code quality

- Built reusable component library and design system, ensuring consistent UI/UX across the platform

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Senior Software Engineer

Tilia | January 2023 - August 2023

- Partnered with cross-functional teams in product and design to revamp critical user-facing features
- Built and shipped work in support of KYC, identity verification, and risk-adjacent tooling as part of Tilia's payments platform (tools team context)
- Spearheaded and guided a team through the successful adoption of Chakra UI, reducing bespoke components and establishing cohesive design patterns
- Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
- Maintained exceptional unit test coverage of over 90% for the Tools team application, alongside comprehensive end-to-end testing
- Led the swift creation and deployment of a homepage reporting dashboard within the initial two months of joining
- Role ended in a company-wide layoff affecting approximately one-third of employees

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Senior Software Engineer

Bloom.wine | September 2020 - October 2022

- Architected a greenfield Next.js application for the primary admin interface of the Bloom product; much of that stack remains in production across many winery clients today
- Engineered Ruby on Rails GraphQL API endpoints shared by multiple applications across mobile and web devices

- Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
- Collaborated with designers using Figma, establishing component library standards usable by both engineering and design
- Integrated DatoCMS and extended application functionality through third-party API integrations (Shopify, Mailchimp, Stripe, Shipstation, ActiveCampaign); all Bloom winery clients were on Shopify Plus

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Software Engineer II

Tune Inc - Marketplace Automation | August 2019 - May 2020

- Built and operated ETL-style data pipelines and integrations supporting marketplace automation workflows
- Automated a client workflow using a Node.js/Express API and React.js, reducing the time to complete the task from 10-15 minutes to 10-30 seconds
- Modernized the primary legacy React.js application by moving from a custom Webpack process to react-scripts, resulting in a 50% reduction in production build size and a 60% reduction in NPM dependencies
- Reduced development machine CPU usage from 75% to <5% by simplifying the build pipeline and dependency graph
- Role ended in a company-wide layoff

Software Developer

Ply Interactive Inc | June 2014 - July 2019

React Native

- Shipped multiple production React Native applications for agency clients—cross-platform mobile apps delivered end to end
- Washington State Fair: React Native app in sustained multi-year production; very high traffic during fair season, from tens of thousands of users into the hundreds of

thousands cumulatively over seasons

Web & platforms

- Developed numerous web applications including Georgetownbeer.com, WashingtonWine.org, TwoWolvesWine.com, Novaerus.com, WellAir, and Plasma-air.com
- Built a standalone Ruby gem Content Management System, enabling simplified content management within Ruby on Rails projects
- Transitioned the agency to modern technologies like GraphQL, Gatsby.js, and React.js, improving delivery times for new projects
- Bloom.wine Product Development: Built extensive reporting features using complex PostgreSQL queries and interactive filtering, saving clients thousands of hours on previously manual tasks
- Engineered a flexible Wine Club membership structure for wineries, allowing for diverse client configuration and broadening the range of potential sales targets
- Communicated and enforced the standards of Shopify's Polaris React.js component library with designers, resulting in efficiencies gained from leveraging pre-built components

COMPANY REFERENCE

Plantfolio (2023-Present) - AI-powered plant management platform with 40,000+ plant database serving both consumer garden management and independent garden center services - plantfolio.garden/

Tilia (2023) - All-in-one payment platform for digital economies - tilia.io/

Bloom.wine (2020-2022) - Commerce services for the alcohol industry; all clients on Shopify Plus - bloom.wine/

Tune Inc - Marketplace Automation (2019-2020) - SaaS platform for managing marketing partnerships across mobile and web - tune.com/

Ply Interactive Inc (2014-2019) - Boutique agency: web and production React Native apps (e.g. Washington State Fair at major scale); developed Bloom.wine product internally (2017-2019)